

Yu Wang

Email: yuw164@ucsd.edu | [Google Scholar](#) | [Homepage](#)

RESEARCH INTERESTS

Equipping LLM with Human-like Memory: Self-Updatable LLMs (Towards LifeSpan Cognitive Systems that can self-update like the human mind);

Multi-Modal Understanding and Generation: Multimodal Memory-Based Understanding (Enabling LLM to hear and see);

EDUCATION

University of California, San Diego

Sept. 2022 ~ present

PhD: Computer Science Engineering

Advisor: Julian McAuley

University of Science and Technology of China

Sept. 2018 ~ July 2022

BS: School of the Gifted Young (Skipped the third year in high school)

Overall GPA: 3.95/4.3 (91.42/100) Ranking: 2/28

Advisor: Xiangnan He

WORK EXPERIENCE

Amazon; Mentor: Yifan Gao;

June. 2023 ~ Dec. 2023

Topic: **MemoryLLM: Towards Self-Updatable Large Language Models**

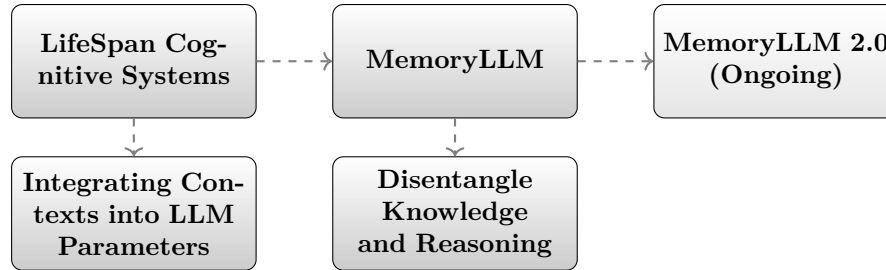
Details: Equipped Large Language Models (LLMs) with an expansive memory pool in latent space, enabling seamless integration of new knowledge. MemoryLLM retains long-term info without performance drops, validated through extensive benchmarks.

MIT-IBM Waston Lab; Mentor: Zexue He;

June. 2024 ~ Sept. 2024

Topic: **MemoryLLM 2.0**

SELECTED PROJECTS AND STRUCTURE



PUBLICATIONS¹

[1] Self-Updatable Large Language Models with Parameter Integration. [PDF](#)

Yu Wang*, Xinshuang Liu*, Xiusi Chen, Sean O'Brien, Junda Wu, Julian McAuley.

[2] Towards LifeSpan Cognitive Systems. [PDF](#)

Yu Wang*, Chi Han*, Tongtong Wu*, Xiaoxin He*, Wangchunshu Zhou, Nafis Sadeq, Xiusi Chen, Zexue He, Wei Wang, Gholamreza Haffari, Heng Ji, Julian McAuley.

[3] CoMMIT: Coordinated Instruction Tuning for Multimodal Large Language Models [PDF](#)

Junda Wu, Xintong Li, Tong Yu, **Yu Wang**, Xiang Chen, Jiuxiang Gu, Lina Yao, Jingbo Shang, Julian McAuley.

[4] Large Scale Knowledge Washing. [PDF](#)

Yu Wang, Ruihan Wu, Zexue He, Xiusi Chen, Julian McAuley.

¹The mark "*" on the names means equal contribution.

- [5] **LVChat: Facilitating Long Video Comprehension.** [PDF](#)
Yu Wang*, Zeyuan Zhang*, Julian McAuley, Zexue He.
- [6] **Outlier Summarization via Human Interpretable Rules.**
 Yuhao Deng, **Yu Wang**, Lei Cao, Lianpeng Qiao, Yu-Ping Wang, Yizhou Yan, Sammual Maddden.
 Accepted to VLDB 2024.
- [7] **Unified Generation, Reconstruction, and Representation: Generalized Diffusion with Adaptive Latent Encoding-Decoding.** [PDF](#)
 Guangyi Liu*, **Yu Wang***, Zeyu Feng*, Qiyu Wu, Liping Tang, Yuan Gao, Zhen Li, Shuguang Cui, Julian McAuley, Eric P. Xing, Zichao Yang, Zhiting Hu.
 Accepted to ICML 2024.
- [8] **MemoryLLM: Towards Self-Updatable Large Language Models.** [PDF](#)
Yu Wang, Yifan Gao, Xiushi Chen, Haoming Jiang, Shiyang Li, Jingfeng Yang, Qingyu Yin, Zheng Li, Xian Li, Bing Yin, Jingbo Shang, Julian McAuley.
 Accepted to ICML 2024.
- [9] **A Fashion Item Recommendation Model in Hyperbolic Space.**
 Ryotaro Shimizu, **Yu Wang**, Masanari Kimura, Yuki Hirakawa, Takashi Wada, Yuki Saito, Julian McAuley.
 Accepted to CVPR 2024 (workshop).
- [10] **Deciphering Compatibility Relationships with Textual Descriptions via Extraction and Explanation.** [PDF](#)
Yu Wang, Zexue He, Zhankui He, Hao Xu, Julian McAuley.
 Accepted to AAAI 2024.
- [11] **Robust and Interpretable Medical Image Classifiers via Concept Bottleneck Models.** [PDF](#)
 An Yan, **Yu Wang**, Petros Karypis, Zexue He, Chengyu Dong, Zihan Wang, Yiwu Zhong, Jingbo Shang, Amilcare Gentili, Chun-Nan Hsu, Julian McAuley.
- [12] **MedEval: A Multi-Level, Multi-Task, and Multi-Domain Medical Benchmark for Language Model Evaluation.** [PDF](#)
 Zexue He, **Yu Wang**, An Yan, Yao Liu, Eric Y Chang, Amilcare Gentili, Julian McAuley, Chun-Nan Hsu.
 Accepted to EMNLP 2023.
- [13] **Learning Concise and Descriptive Attributes for Visual Recognition.** [PDF](#)
 An Yan*, **Yu Wang***, Yiwu Zhong*, Chengyu Dong, Zexue He, Yujie Lu, William Yang Wang, Jingbo Shang, Julian McAuley.
 Accepted to ICCV 2023.
- [14] **Differentiable Invariant Causal Discovery.** [PDF](#)
Yu Wang, An Zhang, Xiang Wang, Xiangnan He, Tat-Seng Chua.
- [15] **Label Denoising through Cross-Model Agreement.** [PDF](#)
Yu Wang, Xin Xin, Zaiqiao Meng, Xiangnan He, Joemon Jose, Fuli Feng.
- [16] **Controlling Bias Exposure for Fair Interpretable Predictions** [PDF](#)
 Zexue He, **Yu Wang**, Julian McAuley, Bodhisattwa Prasad Majumder.
 Accepted to EMNLP 2022 (findings).
- [17] **AutoOD: Automatic Outlier Detection.** [PDF](#)
 Lei Cao, Yizhou Yan, **Yu Wang**, Samuel Madden, Elke A. Rundensteiner.
 Accepted to SIGMOD 2023.
- [18] **Improving Out-of-Distribution Robustness via Selective Augmentation.** [PDF](#)
 Huaxiu Yao*, **Yu Wang***, Sai Li, Weixin Liang, Linjun Zhang, James Zou, Chelsea Finn.
 Accepted to ICML 2022.
- [19] **Learning Robust Recommenders through Cross-Model Agreement.** [PDF](#)
Yu Wang, Xin Xin, Zaiqiao Meng, Xiangnan He, Joemon Jose, Fuli Feng.
 Accepted to WWW 2022.
- [20] **Meta-learning with an Adaptive Task Scheduler.** [PDF](#)
 Huaxiu Yao*, **Yu Wang***, Peilin Zhao, Mehrdad Mahdavi, Defu Lian, Ying Wei, Chelsea Finn
 Accepted to NeurIPS 2021.

AWARDS AND HONORS

- Rose Fund New Lotus Scholarship (For outstanding research achievements). USTC, 2021.
- Baosteel Scholarship (One out of 402 students in School of the Gifted Young). USTC, 2021.
- Rose Fund Public Affairs Scholarship (For active leadership). USTC, 2021.
- Cyrus Tang Foundation Moral Education Scholarship. USTC, 2020.
- Huawei Scholarship (Top 3% of class). USTC, 2020.
- Excellent Student Scholarship – Gold (Top 3% of class). USTC, 2019.
- Provincial first prize in College Mathematics Competition(At most Top 7%). Anhui Province, 2019.